

PROBLEM 1. THE STATIONARY DISTRIBUTION WITH MINIBATCHING.

Consider batched SGD defined by:

$$\Phi_{-} = \eta \hat{g}^B$$

Where $\hat{g}^B$ is the average of B sampled gradients. Let $g = \mathbb{E}[\hat{g}]$ The covariance matrix at batch size B is given by:

$$\Sigma^B = \mathbb{E}[(\hat{g}^B - g)(\hat{g}^{B\top} - g^\top)]$$

The sde model model of SGD is

$$\Phi(t + \Delta t) = \Phi(t) - g\Delta t + \epsilon\sqrt{\Delta t}, \quad \epsilon \sim \mathcal{N}(0, \eta\Sigma^B)$$

Show that for $\eta = B\eta_0$ the stationary distribution is determined by η_0 independent of B .

Proof. We have the following SDE:

$$d\Phi_t = -g(\Phi)dt + \sqrt{\eta}(\Sigma^B)^{1/2}dB_t$$

Let $\mathcal{A}$ be the infinitesimal generator of this SDE. We have:

$$\mathcal{A}^*f = \nabla \cdot (gf) + \frac{1}{2}\nabla^2 \cdot (f\eta\Sigma^B)$$

By forward Kolmogorov, we have:

$$\partial_t p = \mathcal{A}^*p$$

To find the stationary distribution, we set $\partial_t p = 0$ and solve the PDE:

$$\nabla \cdot (gp) + \frac{1}{2}\nabla^2 \cdot (p\eta\Sigma^B) = 0$$

Let's investigate Σ^B . We have:

$$\begin{aligned} \Sigma^B[i, j] &= \mathbb{E}[(\hat{g}^B[i] - g[i])(\hat{g}^B[j] - g[j])] \\ &= \text{Cov}(\hat{g}^B[i], \hat{g}^B[j]) \\ &= \text{Cov}\left(\frac{1}{B} \sum_{b=1}^B g[b, i], \frac{1}{B} \sum_{b=1}^B g[b, j]\right) \\ &= \frac{1}{B^2} \sum_{b=1}^B \sum_{b'=1}^B \text{Cov}(g[b, i], g[b', j]) \\ &= \frac{1}{B^2} \sum_{b=1}^B \text{Cov}(g[b, i], g[b, j]) \end{aligned}$$

Where the last line follows because the samples are independent. Therefore, we have:

$$\Sigma^B[i, j] = \frac{1}{B} \left(\frac{1}{B} \sum_{b=1}^B \text{Cov}(g[b, i], g[b, j]) \right)$$

Let $\sigma_{i,j} = \text{Cov}(g[i], g[j])$. Since the samples are i.i.d. we have $\text{Cov}(g[b, i], g[b, j]) = \sigma_{i,j}$ for all b . Therefore, we have:

$$\Sigma^B[i, j] = \frac{1}{B}\sigma_{i,j}$$

Thus, for all η_0 , we have:

$$\eta\Sigma^B[i, j] = B\eta_0 \frac{1}{B}\sigma_{i,j} = \eta_0\sigma_{i,j}$$

Which has no dependence on B . Thus, in the above PDE, the solution has no dependence on B . □

PROBLEM 2. THE STATIONARY DISTRIBUTION OF WEIGHTED AVERAGE UPDATES.